

Quantitative Reasoning II

(INF 1220, 3 credits)

University of Austin

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Meeting Schedule: 001: MW 11:30-12:45; 002: TR 15:15-16:30; 003: TR 08:30-09:45

Classroom: 001: Scarbrough 311; 002: Scarbrough 354; 003: Scarbrough 311

Office Hours: Refer to course webpage

Course Webpage:
d-dickmann.github.io/QR-II-F26

Course Description

One course in a two-course sequence in quantitative reasoning. Topics include quantifying uncertainty, estimation techniques and applications. Throughout the course examples will be drawn from a variety of fields including physics, biology, and economics; there will be particular emphasis on the laws of nature and analogies among them.

Course Learning Objectives

By the end of this course, students will be able to:

- Quantify uncertainty using probability.
- Conduct a reproducible analysis of real data to answer a research question.
- Differentiate correlation from causation.
- Critique a statistical argument and the evidence behind it.
- Communicate the findings of a data analysis to a non-technical audience.

Course Format

This course is about making sense of the data the world throws at you. It meets twice per week, combining lecture-based instruction with hands-on, computational data analysis. We will move fast and cover many important concepts, pairing conceptual training with practical examples and research questions drawn from a range of fields, including economics and population dynamics. Students complete regular assignments, a midterm, and a final project.

Required Readings

All lecture notes and course materials will be available on the course webpage. Reading assignments will be communicated through the course webpage. Please obtain a copy of the following book that will complement our work in class:

- *Naked Statistics* — Charles Wheelan

Homework

There will be regular take-home assignments. The number, format, and due dates will be posted on the course webpage.

Research Project

The course includes a final project, assessed substantially through an oral exam. Further details will be posted on the course webpage throughout the term.

Topics

Topics covered in this course include:

- Exploratory data analysis and visualization
- Probability, uncertainty, and risk
- Sampling and estimation
- Precision and the uncertainty of estimates
- Statistical inference and the interpretation of research findings
- Correlation, causation, and causal inference
- Regression and prediction
- Reproducible data analysis
- Evaluating statistical claims and arguments
- Communicating quantitative results

Refer to the course webpage for the course schedule.

Evaluation

The course grade will be based on the following components:

- **Homework:** 10%
- **Midterm:** 45%
- **Project:** 40% (includes oral exam)
- **Engagement:** 5% (How often did you participate in class and contribute to class discussion?)

Tools & Technology

Students will need a laptop computer and access to a variety of tools for data analysis (e.g., Python, R, LLM agents). No prior programming experience is required.

Accessibility Statement

Please review the University Accessibility Statement in the student catalog. Students having special needs should contact the Polaris Center or email Accommodations@uaustin.org. Disability Support Services: The University will make reasonable accommodations for students with disabilities in compliance with Section 504 of the Rehabilitation Act and the Americans with Disabilities Act. The purpose of accommodations is to provide equal access to educational opportunities for eligible students with academic and/or physical disabilities.

Attendance Policy

Attendance is mandatory. If you don't show up, you will fail the course. Each student may miss up to 10% of classes for any reason—no excuse required and without penalty—corresponding to 1, 2, or 3 classes in a 1.5, 3.0, or 4.5 credit course, respectively. Any additional absences will result in a final grade penalty of 12% (1.5 credits), 6% (3.0 credits), or 4% (4.5 credits), regardless of the reason. Missing more than 25% of classes—including the allowable absences—without a medical excuse will result in failing the course. Being more than 20 minutes late to class counts as an unapproved absence.

Late Policy

Sometimes we have bad days, bad weeks, and bad terms. In an effort to accommodate any unexpected, unfortunate personal crisis, I have built a grace policy into the course: that is, a one-time, three-day grace period for one homework assignment. You do not have to utilize this policy, but if you find yourself struggling with unexpected personal events, I encourage you to e-mail me as soon as possible to notify me that you are using the grace policy. All other late assignments will be penalized 20 percentage points per day or partial day that they are late. This policy does not apply to the final project.

Intellectual Foundations Learning Objectives

This is an Intellectual Foundations course. The intended goals and outcomes of the Intellectual Foundations Program are as follows:

- Display proficiency in the English language through interpretation of texts, persuasive writing, and effective speaking;
- Model and solve problems using quantitative reasoning tools, and understand how to quantify uncertainty as it applies to decision-making and risk;
- Achieve a critical understanding of the varieties of knowledge, their interrelationship, and the difference between knowledge and wisdom;
- Identify the necessary conditions of flourishing human lives and communities;
- Achieve a critical understanding of modern society and Western culture;
- Understand and appreciate the unique vibrancy of the American form of government and way of life.

Academic Integrity & AI Policy

You may use AI as a tool, but you are responsible for your own work and must be able to explain and defend anything you submit. The midterm and the oral exam are AI-free.